KRAFZEN AI - Internal 60-Day MVP Roadmap (Developer Execution Plan)

Executive Summary

The objective for the next 60 days is to deliver a functional MVP of KRAFZEN AI capable of converting natural voice or text inputs into CAD-ready DXF files. This internal roadmap outlines each developer's responsibilities, KPIs, and weekly milestones. Our stack is finalized: Next.js + Tailwind (Vercel) for frontend, FastAPI (Render) for backend, Supabase for database and auth, and OpenAI Whisper + GPT-4 for AI parsing.

Infrastructure Overview

- **Frontend Hosting:** Vercel (Next.js app)
- **Backend Hosting:** Render (FastAPI service)
- **Database & Auth:** Supabase (PostgreSQL + Auth)
- **AI Services:** Whisper (for transcription) and GPT-4 Turbo (for text parsing)
- **CAD Processing:** ezdxf Python library
- **Monitoring & Analytics:** Sentry, Google Analytics, and Supabase logs
- **CI/CD:** GitHub Actions with auto-deploy to Render and Vercel
- **Version Control:** GitHub

Team Composition & KPIs

Kiruthika Ponnan - Tech Lead / Architect

- Lead backend architecture and deployment on Render.
- Implement CI/CD automation via GitHub Actions.
- Ensure API response time <1s and uptime >99% by demo.
- Deliver architecture documentation and API schema by Week 8.

Sonu - Frontend Lead

- Build full Next.js frontend with Supabase auth and session handling.
- Integrate voice recording, upload, and DXF download flow.
- Ensure responsive design and zero critical UI bugs by demo.
- Coordinate UI/UX review and polish by Week 8.

Krishnarajan V - Full Stack Developer

- Develop backend API routes, CRUD operations, and authentication layers.
- Integrate Supabase database and manage data validation.
- Achieve 100% API test coverage with Postman collections.
- Ensure all routes meet OpenAPI documentation standards.

Laxmi Pranitha Muskula - Data/Systems Analyst

- Design and validate Supabase schema for project, job, and user data.

- Create data dashboards and run integrity validation tests.
- Implement logging and analytics layer via Supabase and GA4.
- Ensure API call tracking and usage reporting is functional by Week 8.

Ashwani Kumar - Jr. Developer / QA Engineer

- Write and maintain test scripts for all FastAPI endpoints.
- Perform regression testing on UI and backend.
- Document QA results and maintain issue tracker.
- Ensure all core features pass regression testing by final demo.

Weekly Milestones (0–60 Days)

Weeks 1–2: Environment Setup & Backend Foundation

- **Kiruthika**: Configure FastAPI backend on Render, set up CI/CD via GitHub Actions.
- **Krishnarajan**: Set up Supabase database schema and connect FastAPI endpoints.
- **Sonu**: Scaffold Next.js project and prepare Vercel deployment pipeline.
- **Laxmi**: Validate Supabase connection and data storage tests.
- **Ashwani**: Test API health checks and setup Postman workspace.

Weeks 3–4: Core AI Integration & API Development

- **Kiruthika**: Implement Whisper transcription service and GPT-4 parsing endpoint.
- **Krishnarajan**: Finalize CRUD endpoints and authentication layers. ● **Sonu**: Integrate voice recording and transcription UI components. ● **Laxmi**: Begin analytics integration for event tracking.
- **Ashwani**: QA test AI endpoints, validate JSON schema parsing.

Weeks 5–6: Frontend Integration & DXF Pipeline

- **Sonu**: Build UI for upload, parse, and DXF generation.
- **Kiruthika**: Implement DXF generator using ezdxf, ensure correct file output.
- **Krishnarajan**: Integrate Supabase storage for project data and DXF files. ● **Laxmi**: Test data flow between frontend and backend through Supabase. ● **Ashwani**: Regression test end-to-end flow (Voice → JSON → DXF).

Weeks 7–8: QA, Stabilization, and Demo Prep

- **Kiruthika**: Finalize backend optimizations, update documentation.
- **Sonu**: Polish UI/UX and ensure all interactions are responsive. ● **Krishnarajan**: Add final API monitoring hooks and error handling. ● **Laxmi**: Validate logs, data analytics, and dashboard metrics.
- **Ashwani**: Perform full regression test suite and create final QA summary.

Definition of Done (Day 60)

By Day 60, the MVP should meet the following criteria:

- Fully functional voice-to-DXF workflow accessible via the web interface.

Backend stable on Render, frontend live on Vercel.
Database and authentication fully managed by Supabase.
All core endpoints documented and passing QA tests.
Error logging active in Sentry and analytics active in GA4.
Demo-ready build available for investors and internal testing.